

AEROSPACE SPATIAL INTELLIGENCE FOR DISASTER RESILIENCE

PRAHARI Orbital Console

Turning satellite data into anticipatory action — from an orbital 3D globe to sub-meter streets, with a Gemini AI advisor and live Google Earth Engine hazard layers.

Anticipatory disaster response · Cyclone-prone coastal India

A prevention platform for a region that pays for disasters after they happen

-  **Problem** Coastal India faces frequent, intensifying cyclones. Hazard data, exposure maps and citizen alerts live in disconnected silos — decisions are made blind at the last mile.
-  **Solution** One continuously-zoomable console fusing live satellite imagery, real storm tracks, surge & flood zones, and infrastructure status — with an AI advisor that answers in plain language.
-  **Opportunity** Governments, insurers and utilities are shifting spend from post-disaster recovery to anticipatory action, parametric insurance and infrastructure hardening.
-  **Why now** Free-tier satellite compute (Earth Engine) and low-cost multimodal AI (Gemini Flash) make sub-meter, explainable risk intelligence deployable with no backend to operate.

The Bay of Bengal is the deadliest cyclone basin on Earth

Majority

of history's deadliest tropical cyclones have formed in the Bay of Bengal

~480 km

of Odisha coastline is directly exposed to landfalling systems

Rising

sea-surface temperatures are intensifying peak wind speeds and surge

Three failures compound at the last mile

● Siloed data

Storm tracks, surge models and vulnerability scores sit in separate tools no responder sees together.

● Blurry maps

Static basemaps stop at district level — you can't see which road, shelter or rooftop actually floods.

● Alerts miss people

Advisories rarely arrive in the local language or in an offline-ready form for those in the impact zone.

A real event that anchors every layer of the platform

185 km/h

Peak sustained winds at landfall near Puri

932 hPa

Central pressure — Extremely Severe Cyclonic Storm

~US\$8.1B

Estimated economic damage across Odisha

1.2 M+

People evacuated in under 48 hours

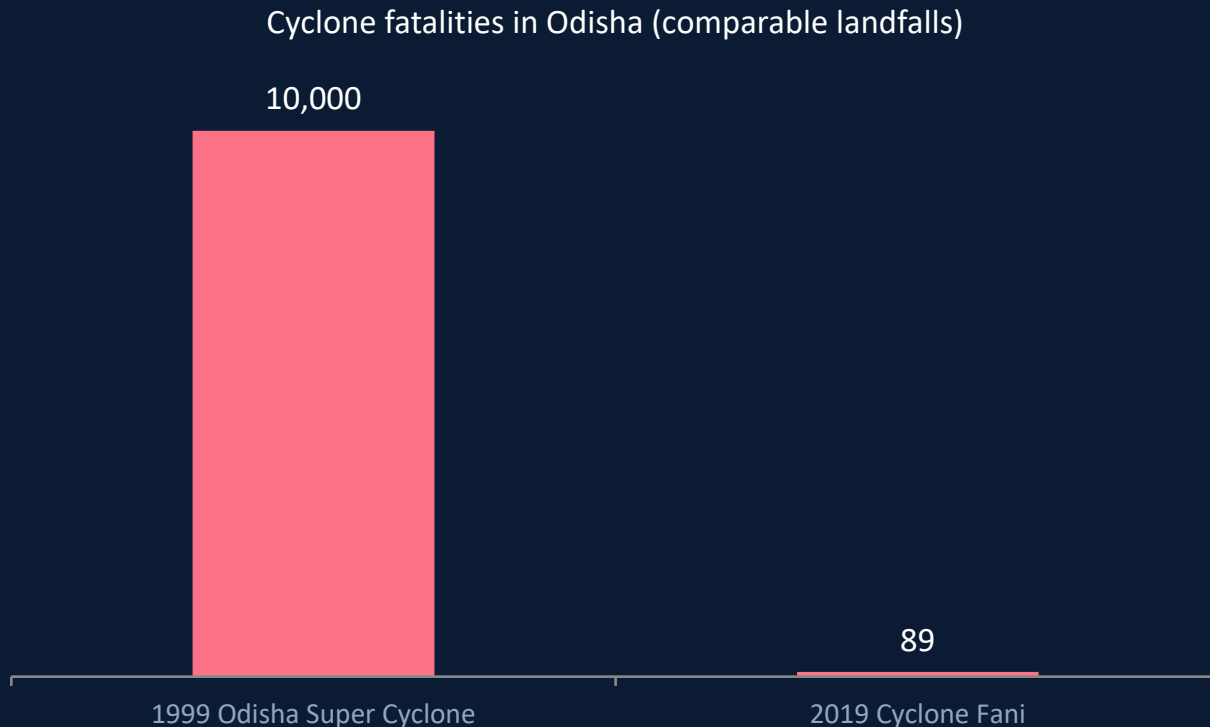
Why Fani, specifically

Fully documented. 13 official best-track waypoints, ward-level vulnerability and shelter status make it an ideal, reproducible test case.

Human-scale stakes. Damage measured in billions and evacuation in millions — small gains in targeting translate to large absolute value.

A validation baseline. Modelled surge and flood extent can be checked against Sentinel-1 SAR observations of the actual inundation.

Early warning already saves thousands of lives — the frontier is assets



What changed

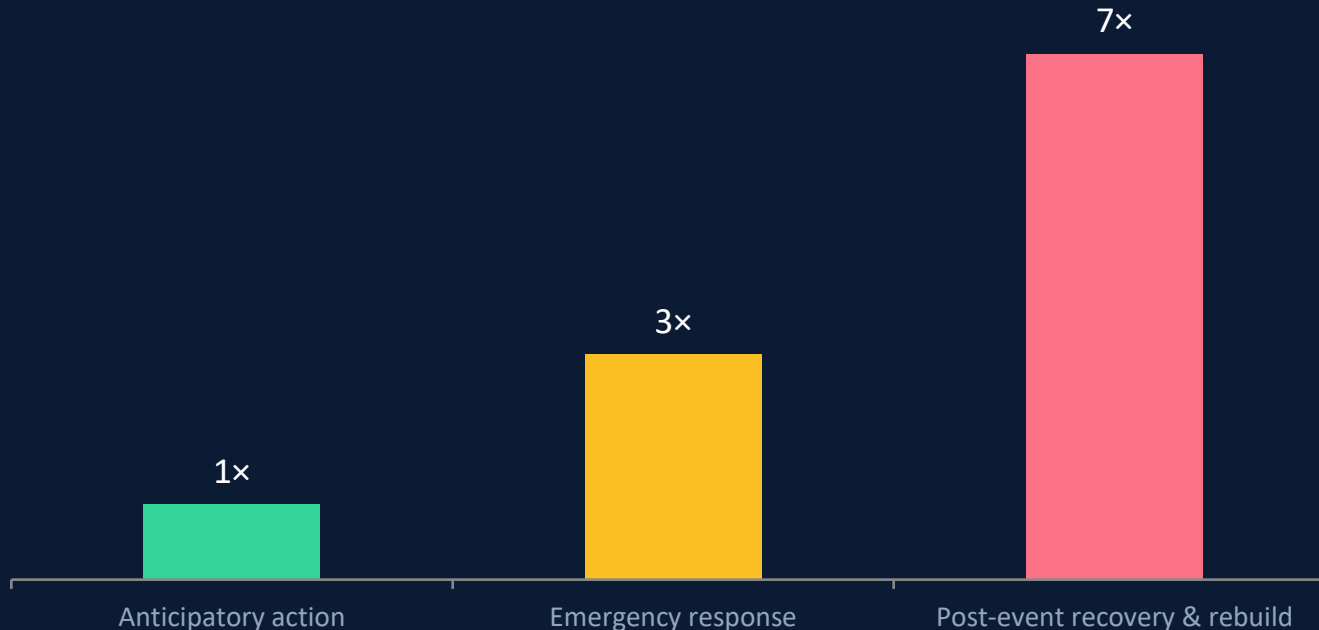
Investment in forecasting, mass evacuation and shelters cut fatalities by **more than 99%** between two comparable landfalls.

The unsolved half: economic loss. Buildings, power, roads and livelihoods still absorb billions. The next decade of value is protecting **assets, not just lives** — exactly where granular, explainable spatial intelligence pays off.

Sources / basis: Fatality figures are widely reported orders of magnitude for the 1999 super cyclone (~10,000) and Fani (89).

Every rupee of anticipatory action displaces several of recovery

Illustrative relative spend to address one major event



● Prevention is cheapest

Targeted evacuation, pre-positioning and hardening cost a fraction of rebuilding after the fact.

● Recovery is where money burns

Reconstruction, lost output and relief dominate the true cost curve of a disaster.

● The lever

Shift budget left on this chart. A platform that improves targeting moves spend from red to green.

Sources / basis: Illustrative ratios for framing only. Direction (prevention < response < recovery) is well established in disaster-economics literature (e.g. GFDRR / World Bank).

A wedge into climate-risk analytics, anchored on public safety

TAM

Global climate & disaster-risk analytics

Insurers, governments, utilities, infrastructure operators worldwide

SAM

Cyclone-exposed emerging coastal economies

South & Southeast Asia disaster-management + parametric insurance

SOM

Beachhead: Indian coastal states

State DMAs, municipal bodies & insurers in Odisha, AP, WB, TN, Gujarat

Entry via government resilience budgets and insurer analytics; expand along the same coastline into utilities and infrastructure.

Sources / basis: Segments are qualitative and directional; no market-size figure is asserted. Sizing to be validated bottom-up with design partners.

One dataset, many buyers along the resilience value chain



The commercial insight

The same fused hazard-and-exposure layer is sold once and monetised many times — as a decision console to authorities, as a risk-scoring feed to insurers, and as a public-good advisory channel to citizens. Multi-sided demand around a single data asset.

Orbit-to-rooftop intelligence in a single browser console



Multi-scale globe

Seamless zoom from a 3D Earth in orbit to sub-meter street tiles — no blurriness, no context loss.



Live hazard layers

Real storm tracks, hydrodynamic surge and flood zones, plus optional Google Earth Engine SAR/DEM overlays.



Gemini AI advisor

Ask in plain language; get grounded, action-oriented answers and drafted trilingual advisories — never invented numbers.



Explainable risk

Ward-level vulnerability scores with the drivers behind them, so officials can defend every evacuation call.

Frontier data and AI, deployable with no backend to operate

Google Earth Engine

- Sentinel-1 SAR flood detection
- Copernicus DEM surge exposure
- WorldPop population exposure
- Petabyte archive, free-tier compute

Gemini Flash AI

- Streaming, low-latency answers
- Grounded on live on-screen data
- Trilingual advisory drafting
- Minimal-thinking cost profile

Web rendering

- MapLibre GL JS v5 globe
- Three.js orbital 3D Earth
- Pure static — any CDN / field laptop
- Client keeps its own credentials

Defensibility grows with proprietary, validated hazard fixtures per coastline and accumulated advisory feedback — not from the models, which anyone can call.

Four live business cases on one platform

● Municipal evacuation

District & city DMAs

Rank wards by real vulnerability, see which shelters are reachable and which roads are severed, and dispatch trilingual orders — before landfall.

● Parametric insurance

Insurers & reinsurers

SAR-verified flood extent and surge depth feed transparent, fast-settling parametric triggers — cutting claims friction and basis risk.

● Infrastructure hardening

Power & road utilities

Map exposure of substations, feeders and arterial roads to prioritise capital hardening and pre-storm shutdown plans.

● Anticipatory financing

Relief agencies & donors

Trigger pre-arranged funds on forecast, not on damage assessment — releasing cash while it still prevents loss.

Three revenue lines around one data asset

Public-sector SaaS

Annual per-state / per-city licences for the decision console, with tiered seats and support. Sticky, budget-backed, renews on the resilience cycle.

Insurance analytics

Usage-based licensing of SAR-verified hazard and exposure feeds to parametric insurers and reinsurers, priced per portfolio or per trigger.

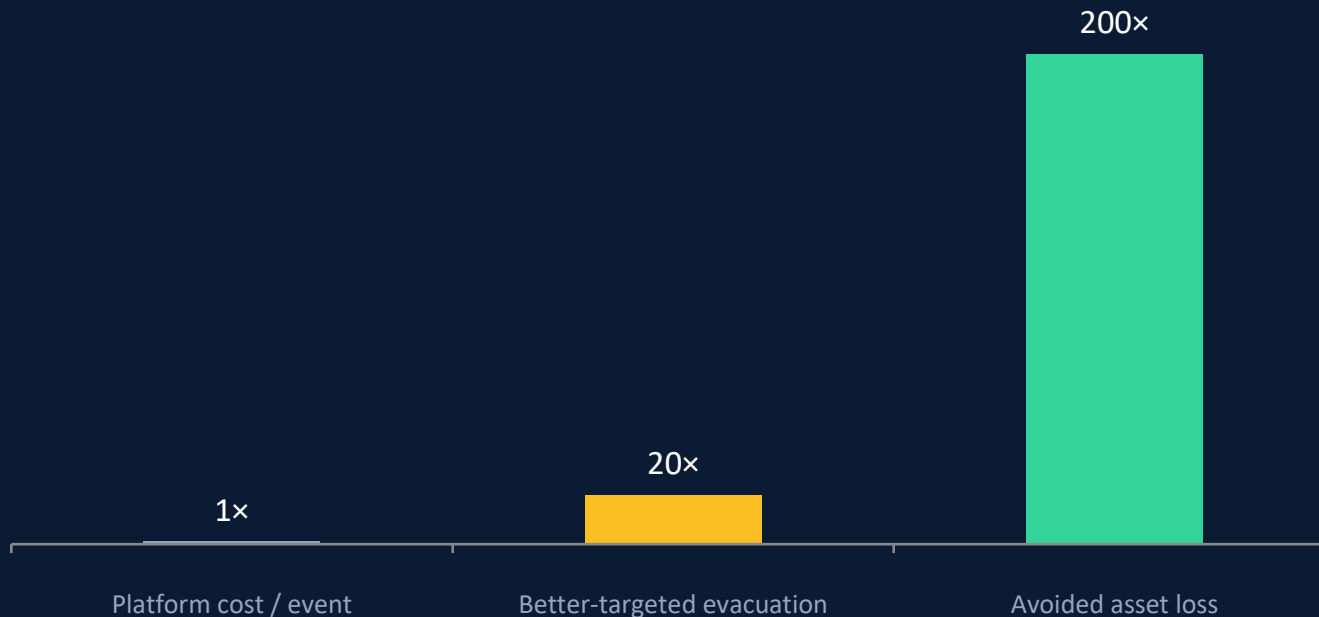
Event advisory

Per-event surge capacity — activation, modelling and advisory dispatch during an active threat — as a high-margin managed service.

Cost advantage: free-tier Earth Engine compute and low-cost Gemini Flash inference mean gross margin scales with coastline coverage, not headcount. No backend to operate keeps delivery lean.

The buyer's ROI is the whole pitch

Illustrative value stack per protected event (indexed)



● Low cost to serve

A subscription is a rounding error against a single event's recovery bill.

● Value in targeting

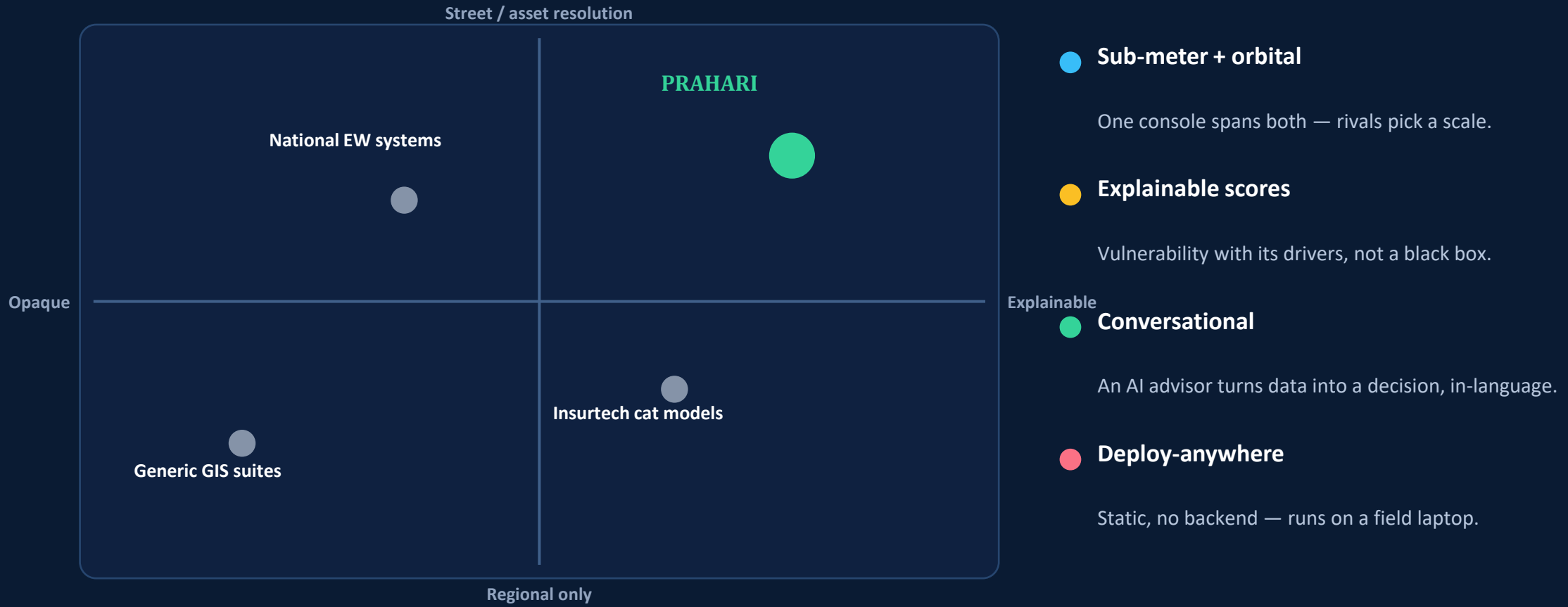
Even a modest improvement in who and what gets protected dwarfs the platform fee.

● Compounding

Every event adds validated fixtures and advisory feedback that sharpen the next forecast.

Sources / basis: Indexed, log-scale illustration for framing the value gap — not a financial projection. Real ROI to be measured with design-partner data.

Granular and explainable, where incumbents are neither



From working prototype to a coastline-scale platform

 **Now**

Prototype

Live console, Gemini advisor, GEE integration on the Fani/Puri case.

 **0–6 mo**

Design partners

1–2 state DMAs + one insurer; validate hazard fixtures on new coastlines.

 **6–18 mo**

Productise

Multi-coast onboarding, parametric feed, SLA-backed event advisory.

 **18 mo+**

Scale

Expand across South & SE Asian coastal economies.

The ask: design-partner access with one state disaster-management authority and one insurer to validate impact and pricing on the next cyclone season.

Sources & assumptions

Cyclone Fani parameters	IMD best-track and post-storm reports; Government of Odisha / OSDMA situation reports. Peak winds ~185 km/h, ~932 hPa, ~US\$8.1B damage, 1.2M+ evacuated are widely reported figures.
Fatality comparison	Orders of magnitude widely reported: ~10,000 (1999 Odisha super cyclone) vs 89 (Fani, 2019).
Basin risk	North Indian Ocean / Bay of Bengal cyclone climatology; NDMA and peer-reviewed literature.
Cost & ROI charts	Illustrative, indexed framing only — not financial projections. Prevention < response < recovery is established in disaster-economics literature (GFDRR / World Bank).
Market sizing	Qualitative and directional. No market-size figure is asserted; to be validated bottom-up with design partners.
Datasets	Sentinel-1 SAR, Copernicus DEM, WorldPop, GPM IMERG via Google Earth Engine; IMD storm tracks.